Relevance-Based Prediction: Factor Selection for Saugus Schools

Four outcomes: MCAS grades 3–8, Dropout Rate, MCAS grade 10 ELA, Education Budget Share

One RBP run per outcome on ALL candidate factors — no pruning (faithful to Kritzman)

Factors split into tiers: Tier 3 (structural) matches Saugus to peer towns; Tiers 1 & 2 (actionable) are ranked

Saugus analyzed as the prediction task; importance scores the actionable factors it can change

Leave-one-out LOO r validates the predictions across all MA districts

Model Summary (all candidates used — no pruning)

MCAS Grades 3–8	Dropout Rate	MCAS Grade 10 (ELA)	Education Budget Share
Predicted on 19 factors	Predicted on 19 factors	Predicted on 19 factors	Predicted on 19 factors
LOO r = 0.90 · r^2 = 0.81	**LOO r = 0.84 · r^2 = 0.70**	**LOO r = 0.90 · r^2 = 0.81**	**LOO r = 0.80 · r^2 = 0.64**
Typical error ± 5.6 pp	Typical error ± 0.7 pp	Typical error ± 6.6 pp	Typical error ± 3.9 pp

LOO r = leave-one-out correlation of predicted vs. actual (range −1 to +1; noise below ≈ 0.15 at this sample size). r^2 = share of town-to-town variance explained. Typical error = mean absolute error, in percentage points.

How to Read This Report — Three Tiers of Factors

This report matches Saugus to demographically-similar towns using Tier 3, then ranks the Tier 1 & 2 factors Saugus can actually change. Structural traits are the backdrop for finding peers — never a recommendation.

TIER 1 — Directly votable (Town Meeting / ballot)

What the town changes by a vote.

- Education's share of the municipal budget
- Residential tax rate · Proposition 2½ override · debt exclusion
- Spending above the state-required school minimum (Ch. 70)
- Reserves (free cash, stabilization) · OPEB funding · capital projects

TIER 2 — Policy / management (administration decides)

What the school department & town administration run day-to-day (funded by Tier 1).

- Chronic absenteeism — attendance & re-engagement programs
- Teacher staffing levels and pay
- How the school dollar splits between classroom and overhead
- Above-minimum school spending funded from reserves

TIER 3 — Structural (NOT changeable by vote)

What the community IS — used ONLY to find Saugus's true peer towns.

- Household income · property wealth · adult education
- Poverty · English-learner · special-education shares
- Housing tenure · district enrollment · regional economy
- → Defines the peer group; never shown as an actionable factor here.

Tiers 1 & 2 = actionable factors (what this report ranks). Tier 3 = structural controls (the peer-matching basis, not recommendations).

How RBP Compares Saugus to Towns Like It — Without Shrinking the Data

It focuses on towns similar to Saugus using the FULL set of MA districts, never a hand-picked subset — so the comparison keeps its statistical power.

1. Full sample, always — no pre-filtering

Every prediction uses all ~170–220 MA districts.

The full sample is what learns which towns are relevant to Saugus (the factor covariance Ω). Hand-filtering to a small peer set first would make that geometry unestimable and turn the importance scores into noise.

2. Signed weighting — contrasts, not exclusion

Each town's weight may be positive OR negative, and the weights sum to 1.

Most-relevant towns get large positive weights; less-relevant towns get small, often negative weights — used as contrasts (subtracted), as in regression. Only ~6–15 of ~170–220 towns sit near zero, and none is excluded.

3. A small effective core, plus a censored cross-check

About half the towns carry small negative weights; ~10–25 carry most of the weight.

RBP also re-predicts from only the top 80% / 50% / 20% most-relevant towns and blends those tight-peer estimates by how reliably each predicts (its fit) — the method decides how much to trust the tight group; no human picks the cutoff.

The math behind it

$$\hat{y}_{Saugus} = \sum_i w_i \, y_i, \qquad \sum_i w_i = 1$$

The prediction is a signed weighted sum of town outcomes; the weights sum to 1.

$$w_i = \frac{1}{N} + \frac{\lambda^2}{n-1}\,(\delta_i \, r_i - \phi\,\bar{r})$$

Each weight = a uniform 1/N baseline + a tilt that turns NEGATIVE below average relevance.

$$r_i = -\frac{1}{2}(x_i - x_S)^T \Omega^{-1}(x_i - x_S) \; + \; \text{info}$$

Relevance r = closeness to Saugus in factor space (Mahalanobis); Ω estimated from all N towns.

$$N_{\text{eff}} = 1 \,/ \sum_i w_i^2$$

Effective # of towns carrying the weight ≈ 10–25 of ~170–220 — a focus, not a deletion.

$$w = x_S^T (X^T X)^{-1} X^T$$

With all factors and no censoring this is exactly OLS — signed weights are inherent to regression.

Because the relevance geometry Ω is learned from the full sample and the weights sum to 1, concentrating the prediction on a relevant core costs no stability — the instability a small hand-picked subset would create.

Combined Model Summary: All Four Outcomes

RBP factor selection results across MCAS grades 3–8, Dropout Rate, MCAS grade 10 ELA, and Education Budget Share

Model	Predicted	Actual	Gap	Direction	LOO r	Factors
MCAS Grades 3–8	23.6%	29.0%	+5.4pp	over	+0.902	19
Dropout Rate	4.1%	2.9%	-1.2pp	over	+0.838	19
MCAS Grade 10 (ELA)	34.9%	47.0%	+12.1pp	over	+0.900	19
Education Budget Share	40.9%	32.3%	-8.6pp	under	+0.801	19

Actionable Factor Cross-Reference — Which Factors Matter Across Outcomes

Actionable factors only (structural traits are matching-only). A factor that matters in multiple outcomes is a robust target.

Factor	# Models	MCAS 3–8	Dropout	MCAS 10	Ed Budget
Students chronically absent (10%+)	4	+0.541	+0.908	+0.622	-0.026
School spend vs. property wealth	4	+0.306	+0.282	+0.442	+0.270
Teacher share of school spending	4	+0.378	+0.327	+0.361	+0.175
Teachers per low-income student	4	+0.375	+0.335	+0.352	+0.127
Teachers per 100 students	4	+0.359	+0.387	+0.346	+0.079
Residential tax rate (per $1,000)	4	+0.322	+0.325	+0.349	+0.149
Spending vs Ch70 required minimum	4	+0.377	+0.314	+0.374	+0.064
Net school spending per pupil	4	+0.326	+0.233	+0.357	+0.047
Ed budget share	3	-0.116	-0.218	-0.163	—
Pension & benefits share of budget	1	—	—	—	+0.393

Green rows = factor matters in 3+ outcomes. Cell colour: green = positive importance, red = negative importance, white = not in that model (—).

What This Means: MCAS Grades 3–8

Predicted 23.6pp · Actual 29.0pp · Gap +5.4pp (higher is better) · Saugus ranks 54 of 219 MA districts on this measure

The Bottom Line

Saugus's MCAS grades 3–8 proficiency (ELA + Math, meeting/exceeding) is 29.0%, 5.4pp better than its demographic prediction of 23.6% — ranking 54 of 219 MA districts, in the top 25%. A district with Saugus's income, poverty, and enrollment profile is predicted to land near this level.

Top Actionable Factors (what Saugus can change)

Highest-importance actionable factors, Saugus vs. statewide median. Structural traits match peers silently and are not shown here.

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	+0.54
Teacher share of school spending	0.349	0.368	+0.38
Spending vs Ch70 required minimum	1.03	1.36	+0.38
Teachers per low-income student	0.126	0.277	+0.37
Teachers per 100 students	5.87	8.07	+0.36

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	19.1%	-12.1pp
Teacher share of school spending	0.349	0.359	+0.010
Spending vs Ch70 required minimum	1.03	1.54	+0.504
Teachers per low-income student	0.126	0.213	+0.087
Teachers per 100 students	5.87	8.81	+2.93

Difference = over-performer median − Saugus. Sign direction depends on the factor; see takeaway below.

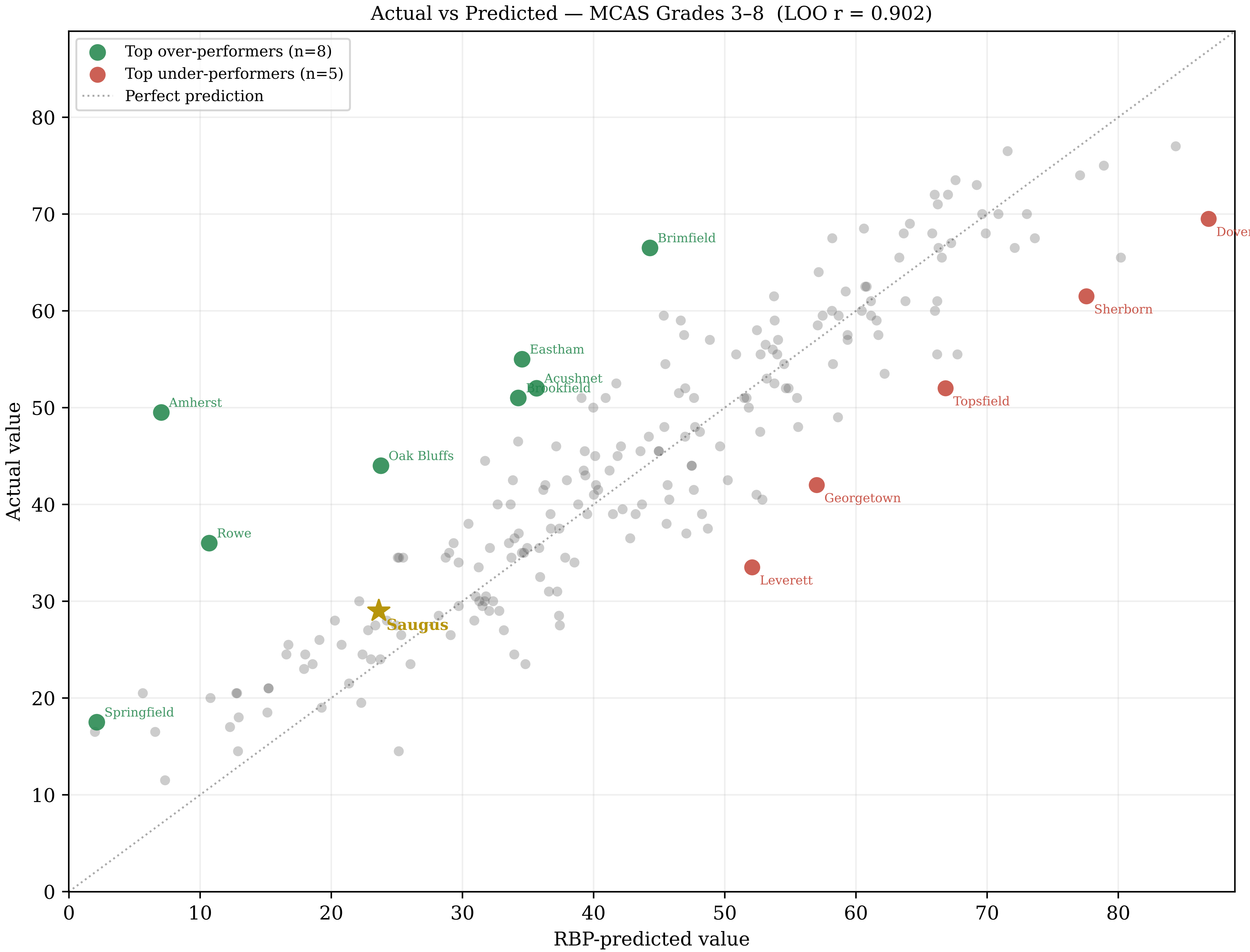
The Takeaway

Saugus does better here than its demographic profile predicts. The strongest actionable factor is students chronically absent (10%+) (31.2% vs the 15.8% state median); the widest gap to the over-performing districts is students chronically absent (10%+) (-12.1pp, peer median − Saugus). (Seed-stability: the top factor is seed-stable; the leading factors below it recur across random grids but their order and importance values jump run-to-run — read them as a co-equal group, not a strict ranking.) The factors below are where Saugus differs most from the towns that beat their own predictions — the most plausible places to move this result.

Left: top actionable factors by RBP importance, Saugus vs. the state median. Right: the same factors for the towns that most beat their own demographic prediction. Structural traits are used for peer-matching only.

Over- and Under-Performers: MCAS Grades 3–8

Saugus gap: +5.4pp · Gold star = Saugus · Green = over-performers · Red = under-performers



Residual = actual − predicted. Positive = performing better than demographics suggest.

What Over-Performers Do Differently: MCAS Grades 3–8

Actionable-factor values for the top over-performing towns vs Saugus — ordered by importance

Actionable-factor values — Saugus (highlighted) vs over-performers
Ordered by |importance|, top 9 shown

Factor	Saugus	Amherst	Rowe	Imp
Students chronically absent (10%+)	31.2	16.7	26.2	+0.541
Teacher share of school spending	0.35	0.31	0.33	+0.378
Spending vs Ch70 required minimum	1.03	1.76	1.91	+0.377
Teachers per low-income student	0.13	0.28	0.20	+0.375
Teachers per 100 students	5.87	9.80	10.6	+0.359
Net school spending per pupil	17,369	31,268	34,298	+0.326
Residential tax rate (per $1,000)	10.7	18.5	5.01	+0.322
School spend vs. property wealth	0.07	0.35	0.03	+0.306
Ed budget share	32.3	52.2	29.9	-0.116

Over-performer residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	29.0	23.6	+5.4pp
Amherst	49.5	7.0	+42.5pp
Rowe	36.0	10.7	+25.3pp
Brimfield	66.5	44.3	+22.2pp
Eastham	55.0	34.5	+20.5pp
Oak Bluffs	44.0	23.8	+20.2pp
Brookfield	51.0	34.3	+16.7pp

Gap = actual − predicted. Positive = outperforming demographic expectation. Factor values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: Dropout Rate

Predicted 4.1pp · Actual 2.9pp · Gap -1.2pp (lower is better) · Saugus ranks 27 of 168 MA districts on this measure

The Bottom Line

Saugus's dropout rate is 2.9%, 1.2pp better than its demographic prediction of 4.1% — ranking 27 of 168 MA districts, in the top 25%. Dropout reflects re-engagement and attendance recovery as much as demographics.

Top Actionable Factors (what Saugus can change)

Highest-importance actionable factors, Saugus vs. statewide median. Structural traits match peers silently and are not shown here.

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	+0.91
Teachers per 100 students	5.87	8.07	+0.39
Teachers per low-income student	0.126	0.277	+0.33
Teacher share of school spending	0.349	0.368	+0.33
Residential tax rate (per $1,000)	10.7	12.6	+0.33

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	24.3%	-6.9pp
Teachers per 100 students	5.87	7.46	+1.59
Teachers per low-income student	0.126	0.133	+0.007
Teacher share of school spending	0.349	0.366	+0.017
Residential tax rate (per $1,000)	10.7	13.4	+2.79

Difference = over-performer median − Saugus. Sign direction depends on the factor; see takeaway below.

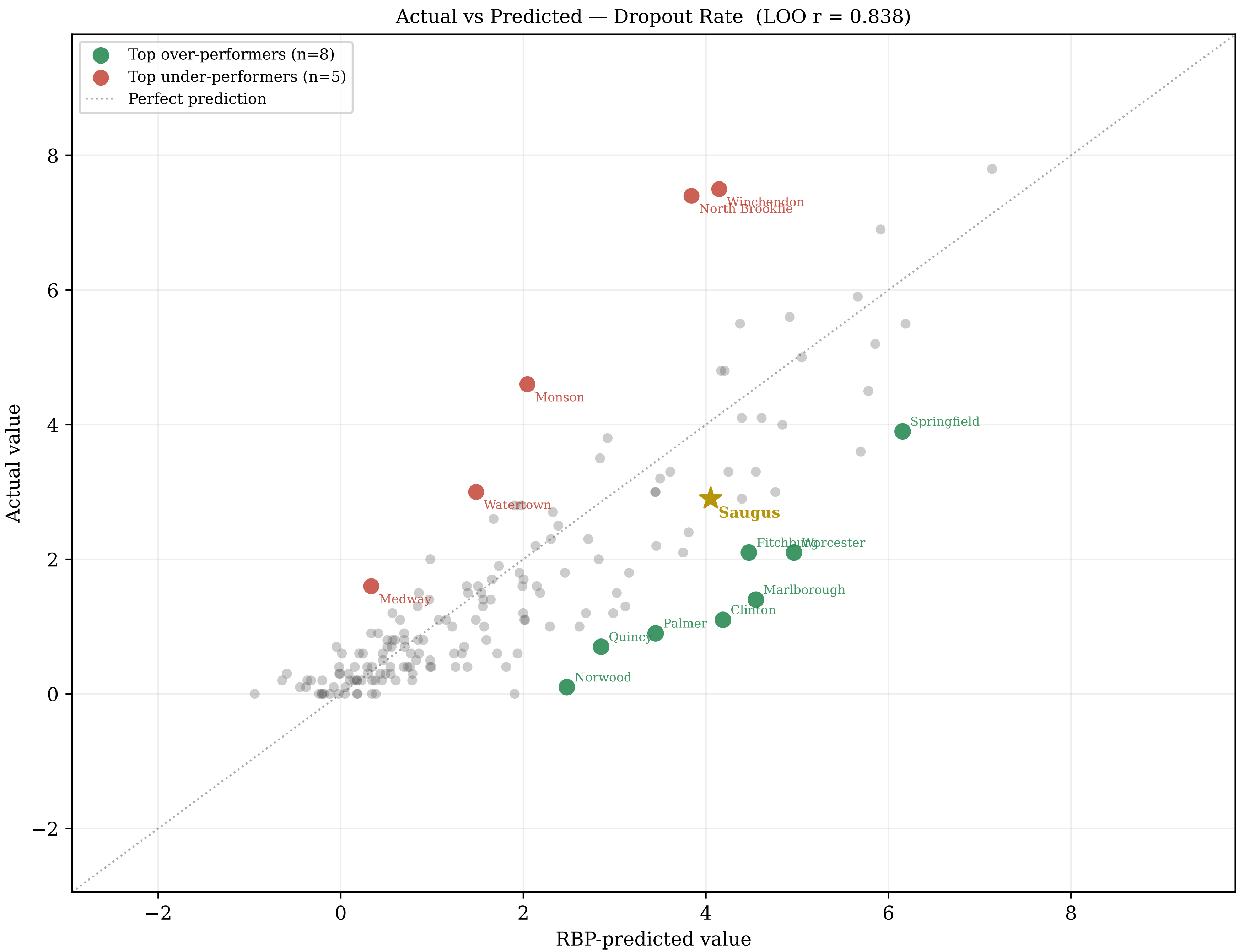
The Takeaway

Saugus does better here than its demographic profile predicts. The strongest actionable factor is students chronically absent (10%+) (31.2% vs the 15.8% state median); the widest gap to the over-performing districts is students chronically absent (10%+) (-6.9pp, peer median − Saugus). (Seed-stability: the top factor is seed-stable; the leading factors below it recur across random grids but their order and importance values jump run-to-run — read them as a co-equal group, not a strict ranking.) Chronic absenteeism is the factor to watch — the main risk to this result if it climbs.

Left: top actionable factors by RBP importance, Saugus vs. the state median. Right: the same factors for the towns that most beat their own demographic prediction. Structural traits are used for peer-matching only.

Over- and Under-Performers: Dropout Rate

Saugus gap: -1.2pp · Gold star = Saugus · Green = over-performers · Red = under-performers



Residual = actual − predicted. Positive = performing better than demographics suggest.

What Over-Performers Do Differently: Dropout Rate

Actionable-factor values for the top over-performing towns vs Saugus — ordered by importance

Actionable-factor values — Saugus (highlighted) vs over-performers
Ordered by |importance|, top 9 shown

Factor	Saugus	Marlborou	Clinton	Imp
Students chronically absent (10%+)	31.2	23.5	7.70	+0.908
Teachers per 100 students	5.87	7.36	7.31	+0.387
Teachers per low-income student	0.13	0.13	0.14	+0.335
Teacher share of school spending	0.35	0.37	0.39	+0.327
Residential tax rate (per $1,000)	10.7	10.2	13.1	+0.325
Spending vs Ch70 required minimum	1.03	1.11	1.08	+0.314
School spend vs. property wealth	0.07	0.09	0.12	+0.282
Net school spending per pupil	17,369	20,169	18,512	+0.233
Ed budget share	32.3	43.0	49.6	-0.218

Over-performer residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	2.9	4.1	-1.2pp
Marlborough	1.4	4.5	+3.1pp better
Clinton	1.1	4.2	+3.1pp better
Worcester	2.1	5.0	+2.9pp better
Palmer	0.9	3.4	+2.5pp better
Norwood	0.1	2.5	+2.4pp better
Fitchburg	2.1	4.5	+2.4pp better

Gap = actual − predicted. Positive = outperforming demographic expectation. Factor values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: MCAS Grade 10 (ELA)

Predicted 34.9pp · Actual 47.0pp · Gap +12.1pp (higher is better) · Saugus ranks 10 of 168 MA districts on this measure

The Bottom Line

Saugus's grade 10 ELA proficiency is 47.0%, 12.1pp better than its demographic prediction of 34.9% — ranking 10 of 168 MA districts, in the top 10% of MA districts. Note: the grade 3–8 academic base is an outcome, not an actionable factor, so it is excluded from this model — part of the over-performance shown here reflects that omitted prior strength rather than a grade-10 effect.

Top Actionable Factors (what Saugus can change)

Highest-importance actionable factors, Saugus vs. statewide median. Structural traits match peers silently and are not shown here.

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	+0.62
School spend vs. property wealth	0.068	0.082	+0.44
Spending vs Ch70 required minimum	1.03	1.36	+0.37
Teacher share of school spending	0.349	0.368	+0.36
Net school spending per pupil	$17,369	$20,865	+0.36

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	18.9%	-12.3pp
School spend vs. property wealth	0.068	0.090	+0.023
Spending vs Ch70 required minimum	1.03	1.40	+0.369
Teacher share of school spending	0.349	0.363	+0.014
Net school spending per pupil	$17,369	$24,272	+6,904

Difference = over-performer median − Saugus. Sign direction depends on the factor; see takeaway below.

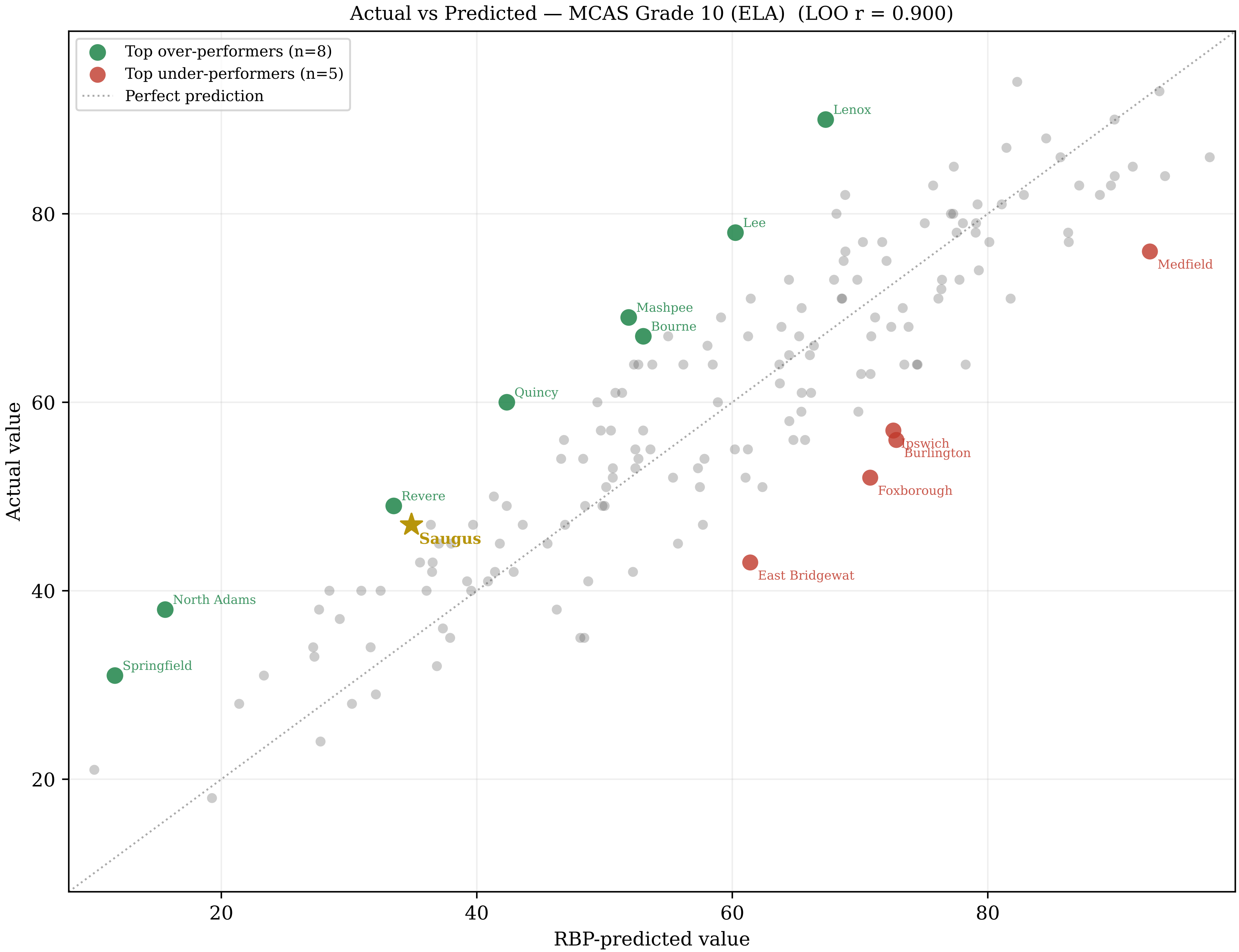
The Takeaway

Saugus does better here than its demographic profile predicts. The strongest actionable factor is students chronically absent (10%+) (31.2% vs the 15.8% state median); the widest gap to the over-performing districts is net school spending per pupil (+6,904, peer median − Saugus). (Seed-stability: the top factor is seed-stable; the leading factors below it recur across random grids but their order and importance values jump run-to-run — read them as a co-equal group, not a strict ranking.) Among the actionable factors, chronic absenteeism and teacher staffing are the most visible — Saugus clears this bar despite lagging peers on both.

Left: top actionable factors by RBP importance, Saugus vs. the state median. Right: the same factors for the towns that most beat their own demographic prediction. Structural traits are used for peer-matching only.

Over- and Under-Performers: MCAS Grade 10 (ELA)

Saugus gap: +12.1pp · Gold star = Saugus · Green = over-performers · Red = under-performers



Residual = actual − predicted. Positive = performing better than demographics suggest.

What Over-Performers Do Differently: MCAS Grade 10 (ELA)

Actionable-factor values for the top over-performing towns vs Saugus — ordered by importance

Actionable-factor values — Saugus (highlighted) vs over-performers
Ordered by |importance|, top 9 shown

Factor	Saugus	Lenox	North Ada	Imp
Students chronically absent (10%+)	31.2	18.3	30.4	+0.622
School spend vs. property wealth	0.07	0.06	0.31	+0.442
Spending vs Ch70 required minimum	1.03	1.54	1.45	+0.374
Teacher share of school spending	0.35	0.37	0.30	+0.361
Net school spending per pupil	17,369	23,945	26,299	+0.357
Teachers per low-income student	0.13	0.41	0.12	+0.352
Residential tax rate (per $1,000)	10.7	9.07	17.1	+0.349
Teachers per 100 students	5.87	10.1	8.67	+0.346
Ed budget share	32.3	48.6	40.3	-0.163

Over-performer residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	47.0	34.9	+12.1pp
Lenox	90.0	67.3	+22.7pp
North Adams	38.0	15.6	+22.4pp
Springfield	31.0	11.7	+19.3pp
Lee	78.0	60.2	+17.8pp
Quincy	60.0	42.3	+17.7pp
Mashpee	69.0	51.9	+17.1pp

Gap = actual − predicted. Positive = outperforming demographic expectation. Factor values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: Education Budget Share

Predicted 40.9pp · Actual 32.3pp · Gap -8.6pp (higher is better) · Saugus ranks 214 of 219 MA districts on this measure

The Bottom Line

Saugus's education budget share is 32.3%, 8.6pp worse than its demographic prediction of 40.9% — ranking 214 of 219 MA districts, in the bottom 10%. Unlike the academic measures, this gap reflects a municipal budgeting choice, not a demographic constraint.

Top Actionable Factors (what Saugus can change)

Highest-importance actionable factors, Saugus vs. statewide median. Structural traits match peers silently and are not shown here.

Factor	Saugus	MA median	Importance
Pension & benefits share of budget	25.3%	17.5%	+0.39
School spend vs. property wealth	0.068	0.082	+0.27
Teacher share of school spending	0.349	0.368	+0.18
Residential tax rate (per $1,000)	10.7	12.6	+0.15
Teachers per low-income student	0.126	0.277	+0.13

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad for the outcome itself.

What Over-Performing Towns Do Differently

Same factors, for the 8 MA districts that most exceed their own prediction:

Factor	Saugus	Peer median	Difference
Pension & benefits share of budget	25.3%	16.1%	-9.2pp
School spend vs. property wealth	0.068	0.137	+0.069
Teacher share of school spending	0.349	0.315	-0.035
Residential tax rate (per $1,000)	10.7	12.5	+1.87
Teachers per low-income student	0.126	0.165	+0.039

Difference = over-performer median − Saugus. Sign direction depends on the factor; see takeaway below.

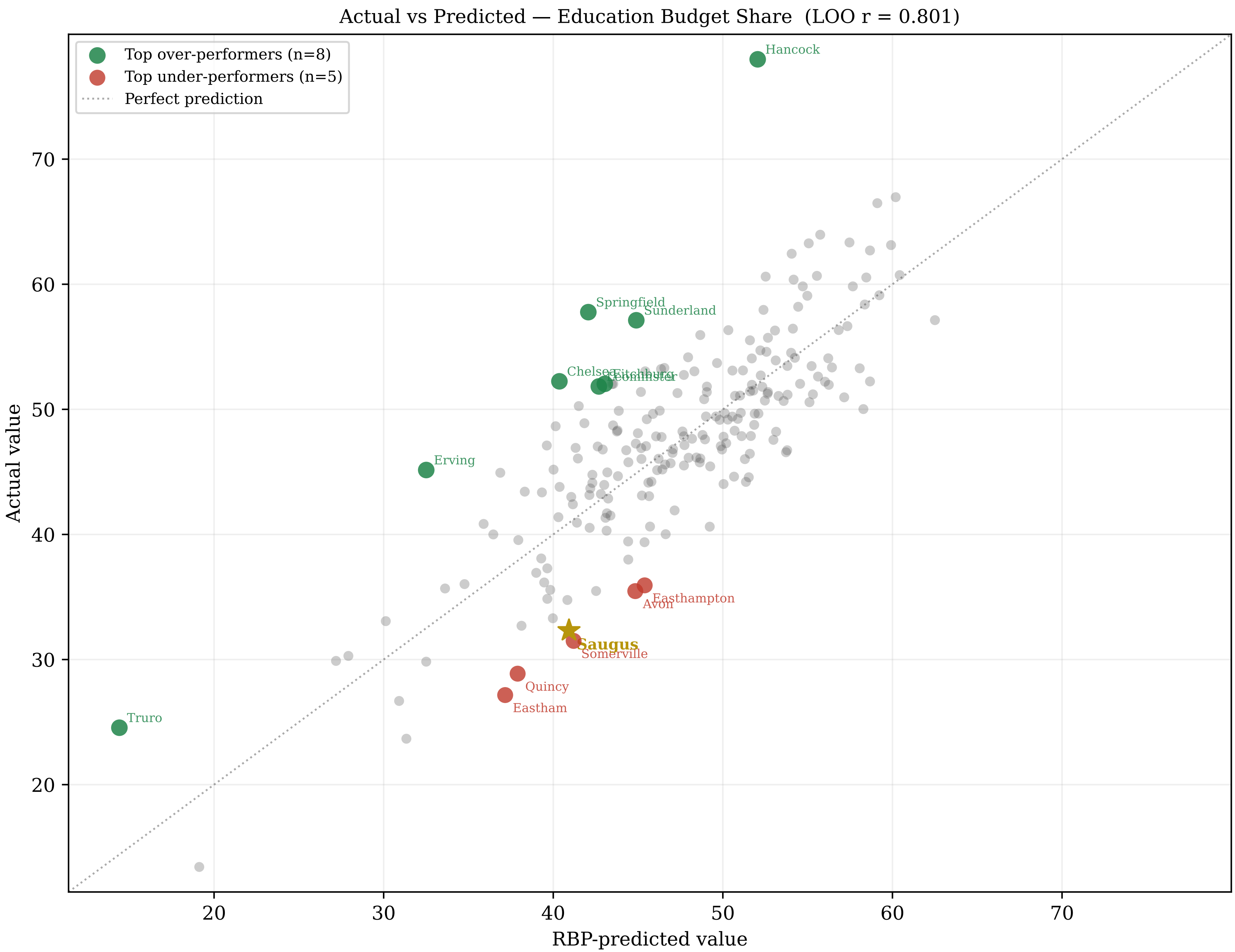
The Takeaway

Saugus does worse here than its demographic profile predicts. The strongest actionable factor is pension & benefits share of budget (25.3% vs the 17.5% state median); the widest gap to the over-performing districts is pension & benefits share of budget (-9.2pp, peer median − Saugus). (Seed-stability: the top factor is seed-stable; the leading factors below it recur across random grids but their order and importance values jump run-to-run — read them as a co-equal group, not a strict ranking.) The top factors are competing budget categories that crowd out the education share — the choice here sits with Town Meeting and municipal budgeting, not the school district.

Left: top actionable factors by RBP importance, Saugus vs. the state median. Right: the same factors for the towns that most beat their own demographic prediction. Structural traits are used for peer-matching only.

Over- and Under-Performers: Education Budget Share

Saugus gap: -8.6pp · Gold star = Saugus · Green = over-performers · Red = under-performers



Residual = actual − predicted. Positive = performing better than demographics suggest.

What Over-Performers Do Differently: Education Budget Share

Actionable-factor values for the top over-performing towns vs Saugus — ordered by importance

Actionable-factor values — Saugus (highlighted) vs over-performers
Ordered by |importance|, top 9 shown

Factor	Saugus	Hancock	Springfie	Imp
Pension & benefits share of budget	25.3	2.43	19.7	+0.393
School spend vs. property wealth	0.07	0.04	0.30	+0.270
Teacher share of school spending	0.35	0.38	0.29	+0.175
Residential tax rate (per $1,000)	10.7	2.79	16.1	+0.149
Teachers per low-income student	0.13	0.20	0.09	+0.127
Teachers per 100 students	5.87	7.36	7.37	+0.079
Spending vs Ch70 required minimum	1.03	1.42	1.26	+0.064
Net school spending per pupil	17,369	21,836	26,616	+0.047
Students chronically absent (10%+)	31.2	16.9	32.4	-0.026

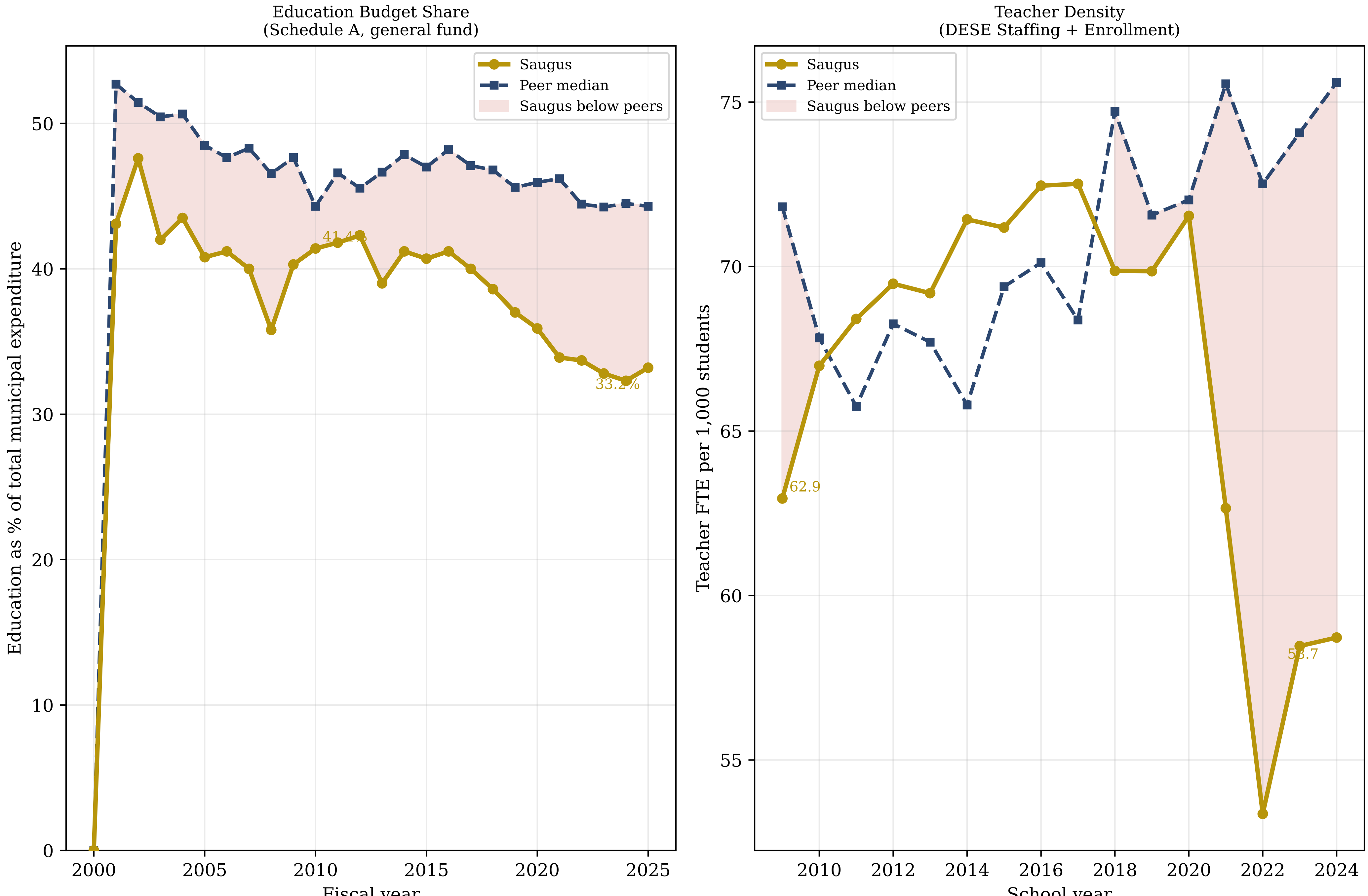
Over-performer residuals vs Saugus

District	Actual	Predicted	Gap
>> Saugus	32.3	40.9	-8.6pp
Hancock	78.0	52.1	+25.9pp
Springfield	57.8	42.1	+15.7pp
Erving	45.1	32.5	+12.6pp
Sunderland	57.1	44.9	+12.2pp
Chelsea	52.2	40.4	+11.9pp
Truro	24.6	14.4	+10.1pp

Gap = actual − predicted. Positive = outperforming demographic expectation. Factor values shown in raw units; importance scores from RBP Exhibit 5.

Budget Allocation & Teacher Density — 15-Year Trajectory

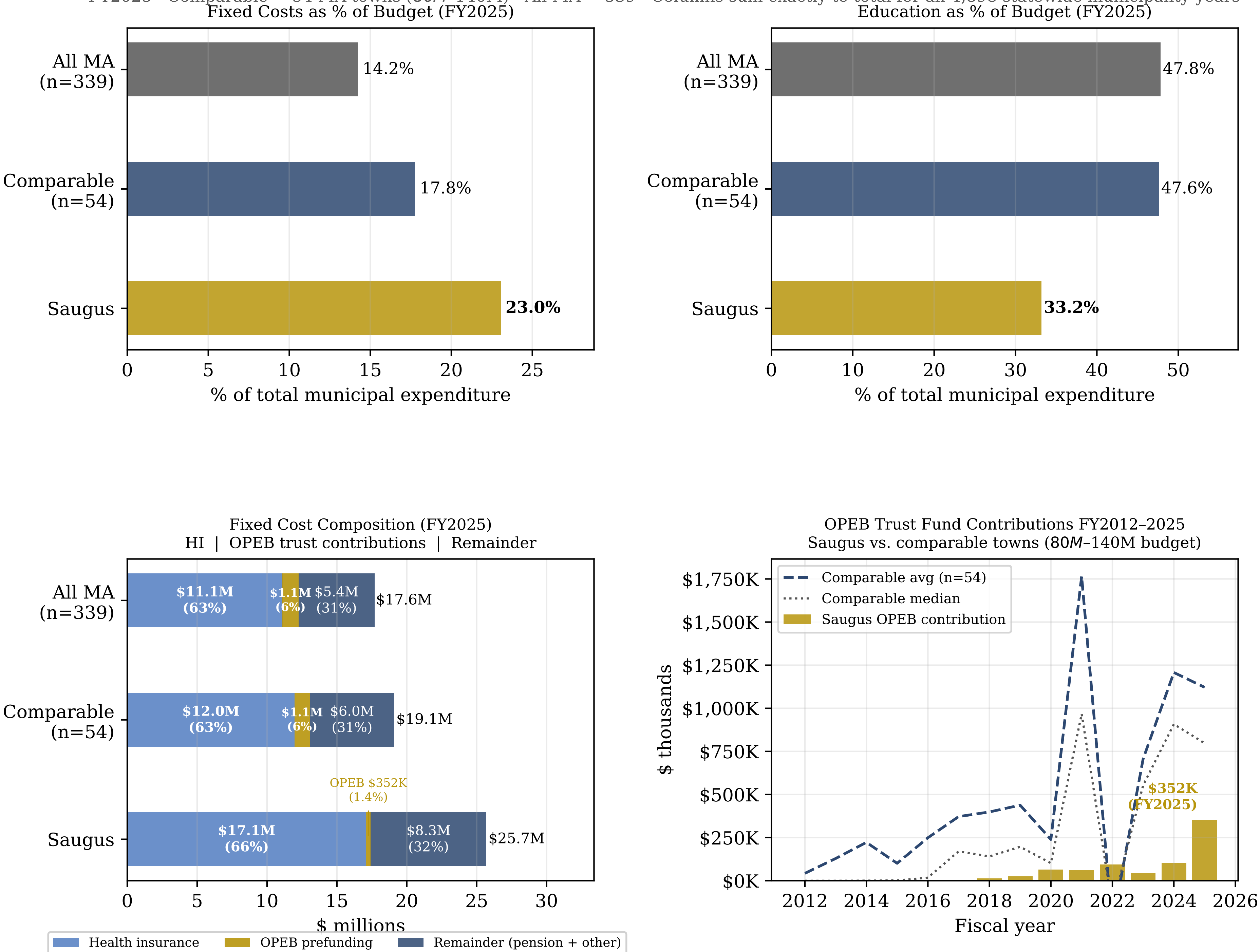
Source: MA DLS Schedule A (budget) · MA DESE Staffing (teachers) · Peer comparison: 8 demographically similar overachiever towns



Independent cross-validation (computed in this run): standardized Ridge regression on 219 MA districts (5-fold CV R^2=0.82) ranks students chronically absent (10%+) #12 of 19 standardized predictors of MCAS 3–8 (|standardized β|=0.08) — a divergence, not corroboration: a linear model assigns students chronically absent (10%+) a small standardized coefficient, whereas RBP's Exhibit 5 ranks it among the top factors — the kind of nonlinear / interaction effect RBP is designed to surface and linear regression misses. Peer set: 8 demographically similar overachiever towns (Mahalanobis d_M ≤ 2).

Fixed Costs: Sources, Scale, and OPEB Prefunding

FY2025 · Comparable = 54 MA towns (80M–140M) · All MA = 339 · Columns sum exactly to total for all 4,898 statewide municipality-years



contributions from Schedule A Part 6 trust fund revenues (municipal_trust_funds, Revenues). Remainder = pension assessment (Essex Regional Retirement System) + workers comp insurance + unemployment + other. Sa

Optimum Profile — What Overachievers Look Like

Each grid shows actionable factor values for Saugus vs over-performer towns. Showing both peer groups tests whether the finding holds across definitions.

Overachievers — all 20 MA districts that beat their demographic prediction

Feature	Saugus	Springfie	North Ada	Fitchburg	Quincy	Amherst	Peer med	Gap	N
Chronic absenteeism (%)	31.2%	32.4%	30.4%	34.0%	16.5%	16.7%	19.9%	-11.3pp	20
Teachers / 100 students	5.87	7.37	8.67	7.59	7.51	9.80	8.30	+2.43	20
Teachers per low-income student	0.126	0.087	0.121	0.105	0.159	0.285	0.199	+0.073	20
Spending vs Ch70 minimum (×)	1.03	1.26	1.45	1.09	1.09	1.76	1.27	+0.238	20
School spending vs. property wealth	0.068	0.300	0.315	0.181	0.086	0.349	0.106	+0.039	20
Teacher share of school $	0.349	0.291	0.303	0.351	0.363	0.312	0.369	+0.020	20

Shown: Springfield, North Adams, Fitchburg, Quincy, Amherst (+ 15 more in median)

Demographically Similar Overachievers — 17 towns within 2σ Mahalanobis distance of Saugus on income, property wealth, poverty, size, education & ELL

Feature	Saugus	Norwood	Clinton	Westport	Lenox	Acushnet	Peer med	Gap	N
Chronic absenteeism (%)	31.2%	25.1%	7.7%	13.8%	18.3%	9.2%	18.3%	-12.9pp	17
Teachers / 100 students	5.87	8.79	7.31	7.46	10.1	6.07	8.38	+2.51	17
Teachers per low-income student	0.126	0.216	0.139	0.216	0.414	0.196	0.202	+0.076	17
Spending vs Ch70 minimum (×)	1.03	1.28	1.08	1.18	1.54	1.11	1.36	+0.323	17
School spending vs. property wealth	0.068	0.081	0.124	0.058	0.063	0.089	0.093	+0.025	17
Teacher share of school $	0.349	0.403	0.385	0.372	0.369	0.371	0.369	+0.020	17

Shown: Norwood, Clinton, Westport, Lenox, Acushnet (+ 12 more in median)

Similarity metric: Mahalanobis distance $d\_M = \sqrt{(\Delta x \cdot \Sigma^{-1} \cdot \Delta x')}$, Σ = sample covariance of all MA districts on (median_hh_income, equalized_income, low_income_pct, total_enrollment, pct_bachelors_plus, ell_pct). Threshold: $d\_M \leq 2$ (2σ). Pre-specified; not tuned to result. Gap = peer median − Saugus. Red = Saugus below peer median, Green = at or above.